

Experiment No. 1

Aim: Create a "Hello Android" App Using Android Studio

Apparatus / Software Required :

Hardware	- Laptop(ASUS Vivobook 16x) - Minimum 8 GB RAM - AMD Ryzen 5 5600H with Radeon Graphics Processor - Internet Connection
Software	- Windows 11 Operating System - Android Studio (Quail) - Java Development Kit (JDK) - Android SDK - Android Emulator (Pixel 7) - Gradle Build System

Theory

Android Studio is the official Integrated Development Environment (IDE) developed by Google for Android application development. It provides tools for designing user interfaces, writing Java/Kotlin code, debugging applications, and testing them using an Android Emulator or a physical Android device.

A **Hello Android** application is the first Android application developed to understand the Android project structure, XML layouts, Java Activity lifecycle, and application execution process. It demonstrates how an Android application is created, built, and executed successfully.

Implementation Steps:

Step 1: Install Android Studio

- Downloaded Android Studio from official website
- Installed with default settings
- Installed SDK, Emulator, and required tools

Step 2: Create New Project

- Opened Android Studio
- Clicked New Project
- Selected Empty Views Activity
- Entered details:
 - Project Name: HelloAndroid
 - Language: Java

- Build: Kotlin DSL
- Clicked Finish

Step 3: Create Emulator

- Opened Device Manager
- Created virtual device (Pixel 7)
- Selected API 34 (Android 14)
- Completed setup

Step 4: Design UI

- Opened activity_main.xml
- Modified default text to "Hello Android!"
- Increased text size
- Made text bold
- Center aligned using ConstraintLayout
- Added another TextView:
 - "Developed by RAJ SIRSODE"
 - Positioned at bottom

Step 5: Run Application

- Started emulator
- Clicked Run
- Application executed successfully

Files Used and Code Changes

1. activity_main.xml

The user interface of the application was designed using the XML layout file activity_main.xml.

Changes Made:

- Modified default text to “Hello Android!”
- Increased text size to 32sp
- Applied bold text style
- Changed text color to red (#FF0000)
- Set background color of layout
- Center aligned the main TextView using ConstraintLayout
- Added a second TextView displaying:
 - “Developed by RAJ SIRSODE”
- Positioned the second TextView at the bottom center

2. MainActivity.java

- Default Java file generated by Android Studio was used
- No additional Java code was required
- The application only involved UI changes

Components Used

Component	Purpose
ConstraintLayout	Used as the parent layout to arrange UI elements
TextView	Displays “Hello Android!” and “Developed by RAJ SIRSODE”

Important XML Attributes Used

Attribute	Description
android:layout_width	Defines width of the view
android:layout_height	Defines height of the view
android:text	Sets the text displayed
android:textSize	Sets size of text
android:textStyle	Makes text bold
android:textColor	Sets text color
android:background	Sets background color
android:fontFamily	Defines font style
layout_constraintTop_toTopOf	Aligns view to top

Important Java Methods Used

1. onCreate()

- It is the first lifecycle method called when the activity starts
- Used to initialize the activity

protected void onCreate(Bundle savedInstanceState)

2. setContentView()

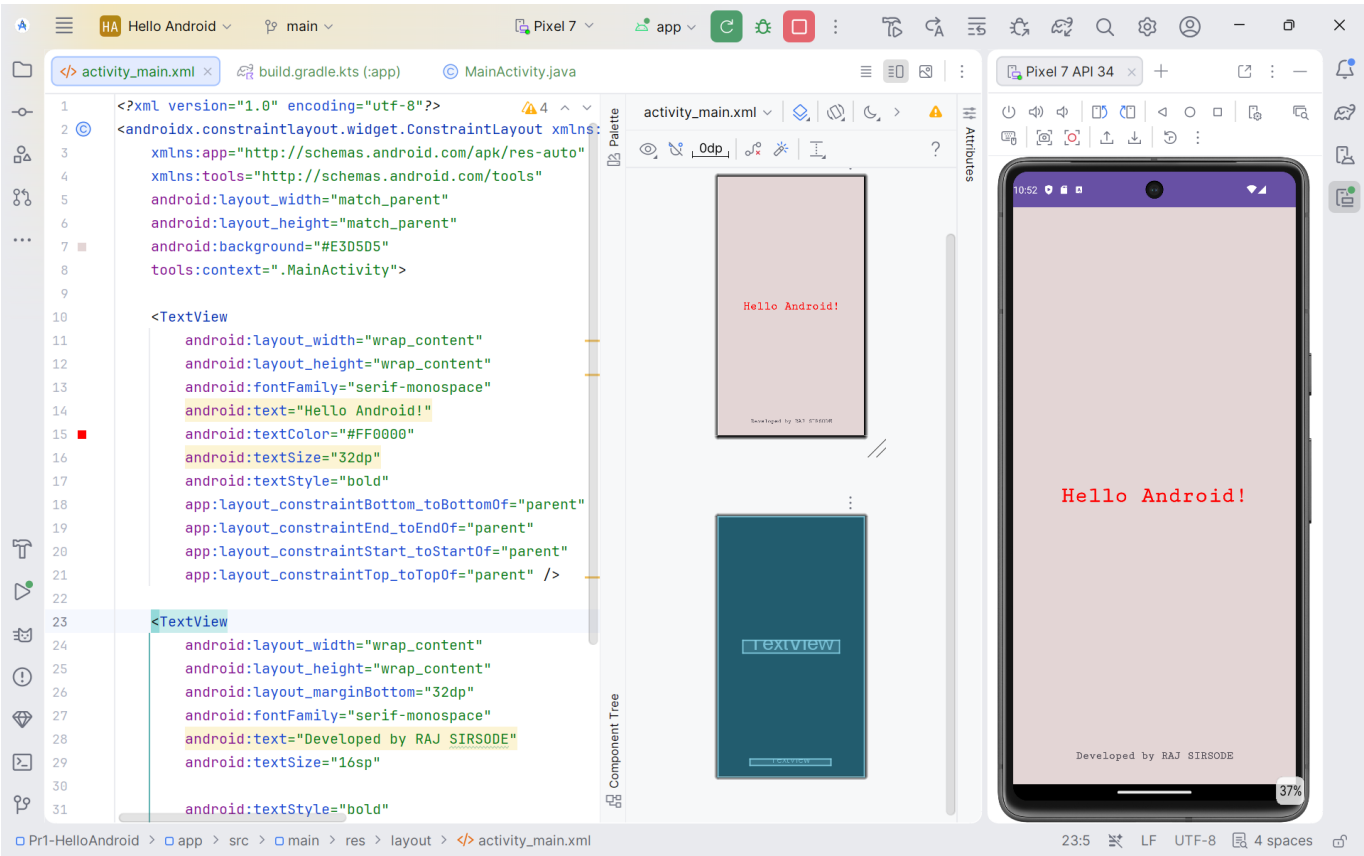
- Used to connect XML layout with Java activity

setContentView(R.layout.activity_main);

Errors Faced and Troubleshooting

Error	Solution
Gradle Sync Failed	Fixed by using compatible dependencies
Incompatible Gradle JVM	Selected JVM 17
New library version error	Downgraded dependency versions
Emulator setup issue	Created Pixel device with API 34
TextView alignment issue	Added proper constraints

Output



Conclusion

The Android development environment was successfully set up using Android Studio. A simple application named **HelloAndroid** was created. The user interface was designed using XML and executed on an Android Emulator. The application successfully displayed the required output, helping to understand the basics of Android app development.

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